GenAI Placement Roadmap

.NET Backend Developer → Applied GenAI / AI Application Engineer

Software Engineer | .NET Backend | Python | GenAI | RAG | LangGraph

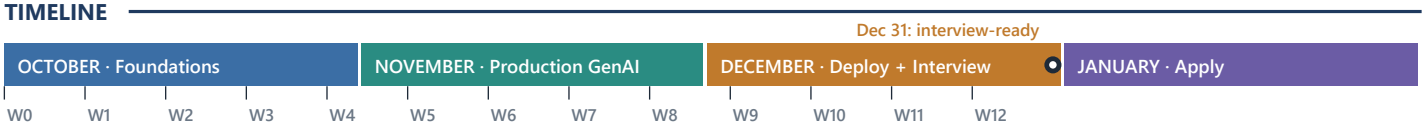
OCT 2026 → JAN 2027

8 hours / day · 13 weeks

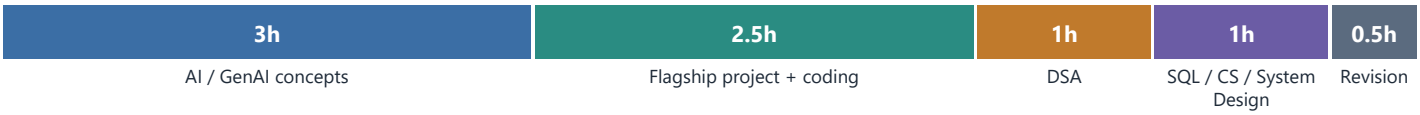
GOAL Interview-ready for GenAI / AI Backend roles
DECEMBER 31, 2026 Done ☐

THEN Start applying for jobs
JANUARY 1, 2027 Done ☐

TIMELINE



DAILY 8-HOUR STUDY SPLIT



Diploma studies run separately, 10 PM – 1/2 AM.

NOW · WEEK 0

WEEK 0 Oct 1 – 7 CURRENT — Done ☐

Python Fundamentals + Flask

- ☐ Data types + operators
- ☐ Lists, tuples, sets, dicts
- ☐ Loops + comprehensions
- ☐ Functions, *args / **kwargs
- ☐ OOP basics
- ☐ Exceptions + file handling
- ☐ Modules, pip, virtual envs
- ☐ Flask routes + JSON APIs

CHECKPOINT by Oct 7 — Done ☐

Ready to start Phase 1

- ☐ Can write small Python scripts without help
- ☐ Built one small Flask REST API
- ☐ Comfortable reading LangChain / LangGraph code
- ☐ Environment set up (uv, .env, Jupyter)

PHASE 1 · OCTOBER — Foundation: NLP · Deep Learning · Transformers · LLM

Oct 8 – Nov 4

WEEK 1 Oct 8 – 14 — Done ☐

NLP

- ☐ Text representation
- ☐ Bag of Words
- ☐ TF-IDF
- ☐ N-grams
- ☐ Tokenization
- ☐ Stemming vs Lemmatization
- ☐ Word Embeddings
- ☐ Word2Vec
- ☐ Cosine Similarity

WEEK 2 Oct 15 – 21 — Done ☐

Deep Learning

- ☐ Neuron
- ☐ Weights and Bias
- ☐ Activation Functions
- ☐ Forward Propagation
- ☐ Loss Function
- ☐ Gradient
- ☐ Backpropagation
- ☐ Optimizers
- ☐ Epoch / Batch
- ☐ Learning Rate
- ☐ Basic CNN
- ☐ Basic RNN
- ☐ Basic LSTM

WEEK 3 Oct 22 – 28 — Done ☐

Transformers

- ☐ Tokens
- ☐ Embeddings
- ☐ Positional Encoding
- ☐ Self-Attention
- ☐ Query / Key / Value
- ☐ Multi-Head Attention
- ☐ Feed Forward Network
- ☐ Layer Normalization
- ☐ Encoder / Decoder
- ☐ BERT
- ☐ GPT
- ☐ T5

WEEK 4 Oct 29 – Nov 4 — Done ☐

LLM Fundamentals

- ☐ Context Window
- ☐ Temperature
- ☐ Top-K
- ☐ Top-P
- ☐ Prompting
- ☐ Structured Output
- ☐ Function / Tool Calling
- ☐ Hallucination
- ☐ RAG
- ☐ Fine-Tuning
- ☐ Agents
- ☐ RAG vs Fine-Tuning vs Agents

PHASE 2 · NOVEMBER — Production GenAI

Nov 5 – Dec 2

WEEK 5 Nov 5 – 11 — Production RAG

Done ☐

- ☐ Document Parsing
- ☐ Cleaning
- ☐ Chunking
- ☐ Embeddings
- ☐ Vector Database
- ☐ Dense Retrieval
- ☐ BM25
- ☐ Hybrid Search
- ☐ Metadata Filtering
- ☐ Query Rewriting
- ☐ Multi-Query Retrieval
- ☐ Reranking
- ☐ Citations

WEEK 6 Nov 12 – 18 — RAG Evaluation

Done ☐

- ☐ Evaluation Dataset
- ☐ Recall@K
- ☐ Precision@K
- ☐ MRR
- ☐ NDCG
- ☐ Context Relevance
- ☐ Answer Relevance
- ☐ Faithfulness
- ☐ LangSmith Tracing
- ☐ LangSmith Evaluation

WEEK 7 Nov 19 – 25 — Agent Engineering

Done ☐

- ☐ LangGraph
- ☐ State
- ☐ Nodes
- ☐ Edges
- ☐ Conditional Edges
- ☐ Reducers
- ☐ Tools
- ☐ Memory
- ☐ Streaming
- ☐ Human-in-the-Loop
- ☐ Routing
- ☐ Supervisor / Worker
- ☐ Reflection
- ☐ Planning
- ☐ When NOT to use multi-agent

WEEK 8 Nov 26 – Dec 2 — MCP + Agents

Done ☐

- ☐ MCP Fundamentals
- ☐ MCP Server
- ☐ MCP Tools
- ☐ MCP Resources
- ☐ MCP Transport
- ☐ STDIO
- ☐ Streamable HTTP
- ☐ MCP + LangGraph Agent
- ☐ Error Handling
- ☐ Permissions / Security

PHASE 3 · DECEMBER — Production · Deployment · Interview Prep

Dec 3 – Dec 30

WEEK 9 Dec 3 – 9 — Python + FastAPI

Done ☐

- ☐ Python OOP
- ☐ Type Hints
- ☐ Decorators
- ☐ Generators
- ☐ Context Managers
- ☐ Async / Await
- ☐ Asyncio
- ☐ Testing
- ☐ FastAPI
- ☐ Pydantic
- ☐ REST APIs
- ☐ JWT
- ☐ Middleware
- ☐ Error Handling
- ☐ Logging

WEEK 10 Dec 10 – 16 — Database + Docker

Done ☐

- ☐ PostgreSQL
- ☐ pgvector
- ☐ Redis
- ☐ Caching
- ☐ Dockerfile
- ☐ Docker Compose
- ☐ Environment Variables
- ☐ Volumes
- ☐ Networks
- ☐ Health Checks

WEEK 11 Dec 17 – 23 — Azure + CI/CD

Done ☐

- ☐ Azure OpenAI
- ☐ Azure Storage
- ☐ Azure AI Search
- ☐ Container Apps / App Service
- ☐ Azure Key Vault
- ☐ Monitoring
- ☐ GitHub Actions
- ☐ CI/CD
- ☐ Docker Deployment

WEEK 12 Dec 24 – 30 — Final Interview Prep

Done ☐

- ☐ Finalize flagship project
- ☐ GitHub README
- ☐ Architecture Diagram
- ☐ Resume
- ☐ AI interview questions
- ☐ RAG interview questions
- ☐ Agent interview questions
- ☐ System Design
- ☐ DSA Revision
- ☐ SQL Revision
- ☐ Mock Interviews

AFTER DECEMBER 15

Do NOT start new frameworks.

BUILD → DEBUG → DEPLOY → EXPLAIN → INTERVIEW

FLAGSHIP PROJECT · ENTERPRISE AI KNOWLEDGE & ACTION ASSISTANT

User → FastAPI → **LangGraph** → **RAG / Agent / MCP** → LLM → Human Approval → Response

- [ ] PDF / DOCX ingestion
- [ ] Chunking
- [ ] Embeddings
- [ ] Vector Search
- [ ] Hybrid Search
- [ ] Metadata Filtering
- [ ] Reranking
- [ ] Citations
- [ ] Query Rewriting
- [ ] LangGraph Agent
- [ ] Tool Calling
- [ ] Memory
- [ ] MCP Server
- [ ] MCP Tools
- [ ] Human-in-the-Loop
- [ ] LangSmith
- [ ] PostgreSQL
- [ ] Redis
- [ ] FastAPI
- [ ] AuthN / AuthZ
- [ ] Docker
- [ ] Azure Deployment
- [ ] CI/CD
- [ ] Prompt-injection awareness

PARALLEL PLACEMENT PREP · DAILY

DSA

50–70 quality problems

- [ ] Arrays
- [ ] Strings
- [ ] Hash Maps
- [ ] Two Pointers
- [ ] Sliding Window
- [ ] Stack
- [ ] Queue
- [ ] Binary Search
- [ ] Linked List
- [ ] Trees
- [ ] Heap
- [ ] Recursion
- [ ] Basic Graphs

SQL

- [ ] JOIN
- [ ] GROUP BY / HAVING
- [ ] Subqueries
- [ ] CTE
- [ ] Window Functions
- [ ] LAG / LEAD
- [ ] ROW_NUMBER
- [ ] RANK / DENSE_RANK
- [ ] CASE
- [ ] UNION
- [ ] EXISTS
- [ ] Date Problems

System Design

- [ ] API Architecture
- [ ] Database Design
- [ ] Vector Search
- [ ] RAG Architecture
- [ ] Agent Architecture
- [ ] Caching
- [ ] Queues
- [ ] Security
- [ ] Monitoring
- [ ] Scalability
- [ ] Cost Optimization

DSA problem tracker — tick one box per solved problem · target 50–70

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JANUARY 2027 · APPLICATION MODE

2h	2h	2h	1h	1h
DSA + SQL	AI interview prep	Project + Sys Design	Job apps	Mock / Revision

Target roles

- [ ] GenAI Developer
- [ ] Junior / Associate GenAI Engineer
- [ ] AI Application Engineer
- [ ] RAG Engineer / Developer
- [ ] Agentic AI Developer
- [ ] AI Backend Engineer
- [ ] .NET + GenAI Developer
- [ ] LLM Application Developer

Your edge: 2 yrs of .NET / backend + GenAI. Combine them, don't discard them.

APIs · SQL · auth · production habits = what most GenAI candidates lack.

DECEMBER 31, 2026

Interview-ready check

- [ ] Concepts clear
- [ ] Flagship project complete
- [ ] RAG evaluated
- [ ] Agent + MCP implemented
- [ ] Dockerized
- [ ] Deployed
- [ ] GitHub ready
- [ ] Resume ready
- [ ] DSA / SQL revised
- [ ] System design practiced
- [ ] Mock interviews completed

> *"Don't try to learn everything. Build deeply, understand what you build, and become interview-ready."*